

Extended

More details: [Empty Quickbooks File Id in PSP](#)

Total Stale Companies on which no Payroll Run activity or Validation call activity happens (means having Empty file Id) : 2,292,068

No Payroll Run & No Validate Subscription Call

548,780 out of 2.2 million customers have never done Validate Subscription call & never ran a single payroll.

No Payroll Run but have validate subscription call

1,743,288 out of 2.2 million customers have done Validate Subscription call but never ran a single payroll.

Only 532,064 out of 1.74 Million customers have done validate subscription call in the last 45 days.

Only 695,385 out of 1.74 Million customers have done validate subscription call in the last 90 days.

Only 948,050 out of 1.74 Million customers have done validate subscription call in the last 180 days.

Only 1,117,115 out of 1.74 Million customers have done validate subscription call in the last 365 days.

Data Analysis

Here are the data analysis for various categories to be targeted for Archiving & Purging.

IOP & QBOE companies

Source System	Companies Count	Query (as of Jan'22)
QBOE	31,203	Query select count(DISTINCT c.company_seq), c.source_system_cd FROM PSPADM. PSP_COMPANY C group by c.source_system_cd;
IOP	628,690	
QBDT	6,335,916	

Active Vs Inactive companies

Inactive company means a company which has not any Entitlement enabled & Entitlement Unit activated.

Status (as of Jan'22)	Company	Employees	Paychecks
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Active	3.2 M Query <pre> SELECT /*+ PARALLEL(8) */ (CASE WHEN ASSET_ITEM_CD IN ('Assisted', 'AssistedAdvantage') THEN 'Assisted' WHEN ASSET_ITEM_CD IN ('DIY', 'DIYDiskDelivery') AND IS_USAGE_BILLING = 0 THEN 'DIY LEGACY' WHEN ASSET_ITEM_CD IN ('DIY', 'DIYDiskDelivery') AND IS_USAGE_BILLING = 1 THEN 'DIY SYMPHONY' ELSE 'OTHERS' END) SUBSCRIPTION_TYPE, count(DISTINCT company_seq) FROM PSPADM.PSP_COMPANY C, PSPADM.PSP_ENTITLEMENT E, PSPADM.PSP_ENTITLEMENT_UNIT EU, PSPADM.PSP_ENTITLEMENT_CODE EC WHERE C.COMPANY_SEQ = EU.COMPANY_FK AND E.ENTITLEMENT_SEQ = EU. ENTITLEMENT_FK AND E.ENTITLEMENT_CODE_FK = EC. ENTITLEMENT_CODE_SEQ AND E.ENTITLEMENT_STATE = 'Enabled' AND EU.ENTITLEMENT_UNIT_STATUS IN ('ActivationHold', 'Activated', 'PendingActivation', 'PendingReactivation', 'ErrorActivating') GROUP BY (CASE WHEN ASSET_ITEM_CD IN ('Assisted', 'AssistedAdvantage') THEN 'Assisted' WHEN ASSET_ITEM_CD IN ('DIY', 'DIYDiskDelivery') AND IS_USAGE_BILLING = 0 THEN 'DIY LEGACY' WHEN ASSET_ITEM_CD IN ('DIY', 'DIYDiskDelivery') AND IS_USAGE_BILLING = 1 THEN 'DIY SYMPHONY' ELSE 'OTHERS' END); </pre>
37 M	37 M Query <pre> select count(*) from (select /*+ PARALLEL (8) */ distinct com.COMPANY_SEQ, emp.employee_seq from pspadm. PSP_COMPANY com, PSPADM. PSP_EMPLOYEE emp, pspadm. PSP_ENTITLEMENT_UNIT eu, PSPADM. PSP_ENTITLEMENT ET, PSPADM. PSP_COMPANY_SERVICE cs where emp.COMPANY_FK = com.COMPANY_SEQ and com. COMPANY_SEQ = eu. company_fk and EU. ENTITLEMENT_FK = ET. ENTITLEMENT_SEQ and ET. ENTITLEMENT_STATE = 'Enabled' and cs.COMPANY_FK = com.COMPANY_SEQ and EU. ENTITLEMENT_UNIT_STATUS in ('Activated', 'PendingActivation', 'PendingReactivation')); </pre>
1.1 B	1.1 B Query <pre> select count(*) from (select /*+ PARALLEL (8) */ distinct com.COMPANY_SEQ, paycheck.paycheck_seq from pspadm. PSP_COMPANY com, PSPADM. PSP_PAYCHECK paycheck, pspadm. PSP_ENTITLEMENT_UNIT eu, PSPADM. PSP_ENTITLEMENT ET, PSPADM. PSP_COMPANY_SERVICE cs where paycheck. COMPANY_FK = com. COMPANY_SEQ and com. COMPANY_SEQ = eu. company_fk and EU. ENTITLEMENT_FK = ET. ENTITLEMENT_SEQ and ET. ENTITLEMENT_STATE = 'Enabled' and cs.COMPANY_FK = com.COMPANY_SEQ and EU. ENTITLEMENT_UNIT_STATUS in ('Activated', 'PendingActivation', 'PendingReactivation')); </pre>

Inactive	1.3 M	9.3 M	533 M
	Query <pre> SELECT /*+ PARALLEL(8) */ (CASE WHEN ASSET_ITEM_CD IN ('Assisted', 'AssistedAdvantage') THEN 'Assisted' WHEN ASSET_ITEM_CD IN ('DIY', 'DIYDiskDelivery') AND IS_USAGE_BILLING = 0 THEN 'DIY LEGACY' WHEN ASSET_ITEM_CD IN ('DIY', 'DIYDiskDelivery') AND IS_USAGE_BILLING = 1 THEN 'DIY SYMPHONY' ELSE 'OTHERS' END) SUBSCRIPTION_TYPE, count(DISTINCT company_seq) FROM PSPADM.PSP_COMPANY C, PSPADM.PSP_ENTITLEMENT E, PSPADM.PSP_ENTITLEMENT_UNIT EU, PSPADM.PSP_ENTITLEMENT_CODE EC WHERE C.COMPANY_SEQ = EU.COMPANY_FK AND E.ENTITLEMENT_SEQ = EU. ENTITLEMENT_FK AND E.ENTITLEMENT_CODE_FK = EC. ENTITLEMENT_CODE_SEQ AND E.ENTITLEMENT_STATE = 'Disabled' AND EU.ENTITLEMENT_UNIT_STATUS NOT IN ('ActivationHold', 'Activated', 'PendingActivation', 'PendingReactivation', 'ErrorActivating') GROUP BY (CASE WHEN ASSET_ITEM_CD IN ('Assisted', 'AssistedAdvantage') THEN 'Assisted' WHEN ASSET_ITEM_CD IN ('DIY', 'DIYDiskDelivery') AND IS_USAGE_BILLING = 0 THEN 'DIY LEGACY' WHEN ASSET_ITEM_CD IN ('DIY', 'DIYDiskDelivery') AND IS_USAGE_BILLING = 1 THEN 'DIY SYMPHONY' ELSE 'OTHERS' END); </pre>	Query <pre> select count(*) from (select /*+ PARALLEL (8) */ distinct com.COMPANY_SEQ, emp.employee_seq from pspadm. PSP_COMPANY com, PSPADM. PSP_EMPLOYEE emp, pspdm. PSP_ENTITLEMENT_UNIT eu, PSPADM. PSP_ENTITLEMENT ET, PSPADM. PSP_COMPANY_SERVICE cs where emp.COMPANY_FK = com.COMPANY_SEQ and com. COMPANY_SEQ = eu. company_fk and EU. ENTITLEMENT_FK = ET. ENTITLEMENT_SEQ and ET. ENTITLEMENT_STATE = 'Disabled' and cs.COMPANY_FK = com.COMPANY_SEQ and EU. ENTITLEMENT_UNIT_STATUS NOT in ('Activated', 'PendingActivation', 'PendingReactivation')); </pre>	Query <pre> select count(*) from (select /*+ PARALLEL (8) */ distinct com.COMPANY_SEQ, paycheck.paycheck_seq from pspadm. PSP_COMPANY com, PSPADM. PSP_PAYCHECK paycheck, pspdm. PSP_ENTITLEMENT_UNIT eu, PSPADM. PSP_ENTITLEMENT ET, PSPADM. PSP_COMPANY_SERVICE cs where paycheck. COMPANY_FK = com. COMPANY_SEQ and com. COMPANY_SEQ = eu. company_fk and EU. ENTITLEMENT_FK = ET. ENTITLEMENT_SEQ and ET. ENTITLEMENT_STATE = 'Disabled' and cs.COMPANY_FK = com.COMPANY_SEQ and EU. ENTITLEMENT_UNIT_STATUS NOT in ('Activated', 'PendingActivation', 'PendingReactivation')); </pre>

Phase 1: Purging non-QBDT companies

As per above data analysis, there are **~660K companies (i.e 9% of total Companies in the PSP DB)** from IOP & QBOE source which are not used any more and kept idle. So targeted these companies to be purged from main Database.

IOP Purging

Why IOP data in the PSP DB: Initially Risk platform capability was not available IOP and so PSP was used to do risk assessment for IOP companies. There was a batch job created on PSP which does risk assessment for IOP companies based on these data.

Current Status: Risk platform is available with IOP and so the job is running any more running, hence the IOP companies data is not more used and can be deleted.

More details on IOP purging: [IOP Data Purging](#)

Execution Strategy

<ToDo>

QBOE Purging

Why QBOE data : Initially there was no infrastructure for QB Online products and so idea was to use the PSP database for that as well and started with that.

Current Status: There are good infrastructure ready for online products and so not using PSP DB anymore and so in the past around 2018, there was an attempt to delete this data But couldn't continue because finance team was using company accounting data for there Account reconciliation.

So, need to check again with finance team about the uses for this data and if at all needed, need to build strategy of how to support account reconciliation and purging the data.

More details about QBOE companies data: [QBOE Data Purging](#)

Execution Strategy

<ToDo>

Phase 2: Archiving Inactive Companies

Per above data, the majority of data is consumed by terminated companies, archiving them to a separate database will improve the performance & maintainability of the main database size at a healthy level.

The strategy is to target such companies which rarely be called for Restore to avoid complexities, operation cost etc. Means such companies should not considered for reactivation again or in very rare scenarios it will be restored back.

Targeted Inactive Companies

Inactive company means: The Company with Entitlement in '**Disabled**' state and all its Entitlement Unit is in ('*ErrorDeactivating*', '*Deactivated*', '*DeactivationHold*', '*Historic*') will be considered as Inactive company.

Targeted Companies: All Inactive Companies which has been inactivated from last 3 & more years because there will be very very rare chance of getting reactive.

<ToDo>: Verify *ErrorDeactivating* & *DeactivationHold* Status.

Data Archival Strategy

Archiving the old data will simplify the migration process.

- **Inactive customers data :** Around 1.17 million customers have not used the product in the last 12 months - [Inactive Customers](#).
- **Cleanup 7 years old data :** As per US compliance laws, financial data older than 7 years can be cleared.

Moving the old data to archival database can significantly reduce database size which will simplify the migration process. PSP's yearly data growth captured here [PSP Yearly Data Growth Distribution](#).